Engineering 54 (2025) 115–126

Contents lists available at ScienceDirect

Engineering

journal homepage: www.elsevier.com/locate/eng

Research
Theory and Key Technologies in Satellite Internet Networking—Article

SatFed: A Resource-Efficient LEO-Satellite-Assisted Heterogeneous Federated Learning Framework

Yuxin Zhang [a,b], Zheng Lin [a,c], Zhe Chen [a], Zihan Fang [a,d], Xianhao Chen [c], Wenjun Zhu [a,b], Jin Zhao [a,b], Yue Gao [a,*]

[a] Institute of Space Internet, Fudan University, Shanghai 200438, China
[b] College of Computer Science and Artificial Intelligence, Fudan University, Shanghai 200438, China
[c] Department of Electrical and Electronic Engineering, University of Hong Kong, Hong Kong 999077, China
[d] Department of Computer Science, City University of Hong Kong, Hong Kong 999077, China

ARTICLE INFO

Article history:
Received 20 September 2024
Revised 17 June 2025
Accepted 14 July 2025
Available online 24 July 2025

Keywords:
Low-Earth-orbit satellite networks
Distributed machine learning
Federated learning
System heterogeneity

ABSTRACT

Traditional federated learning (FL) frameworks rely heavily on terrestrial networks, whose coverage limitations and increasing bandwidth congestion significantly hinder model convergence. Fortunately, the advancement of low-Earth-orbit (LEO) satellite networks offers promising new communication avenues to augment traditional terrestrial FL. Despite this potential, the limited satellite–ground communication bandwidth and the heterogeneous operating environments of ground devices—including variations in data, bandwidth, and computing power—pose substantial challenges for effective and robust satellite-assisted FL. To address these challenges, we propose SatFed, a resource-efficient satellite-assisted heterogeneous FL framework. SatFed implements freshness-based model-prioritization queues to optimize the use of highly constrained satellite–ground bandwidth, ensuring the transmission of the most critical models. Additionally, a multigraph is constructed to capture the real-time heterogeneous relationships between devices, including data distribution, terrestrial bandwidth, and computing capability. This multigraph enables SatFed to aggregate satellite-transmitted models into peer guidance, improving local training in heterogeneous environments. Extensive experiments with real-world LEO satellite networks demonstrate that SatFed achieves superior performance and robustness compared with state-of-the-art benchmarks.

© 2025 THE AUTHORS. Published by Elsevier LTD on behalf of Chinese Academy of Engineering and Higher Education Press Limited Company. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

As edge devices such as drones, smart cameras, and Internet of Things (IoT) sensors proliferate, they generate vast amounts of data at the network edge [1]. This data drives significant advancements in key domains such as video analytics [2], resource management [3], and autonomous driving [4] through machine learning (ML). However, privacy considerations [5] and bandwidth constraints [6] render the transferral of large volumes of data to a cloud server impractical, necessitating the adoption of distributed ML training methods.

To address these challenges, federated learning (FL) [7] has emerged as a promising distributed ML framework that allows edge devices to collaboratively train models without sharing local data. FL typically employs a parameter server (PS) architecture, in which a central server aggregates model updates from participating devices. The efficiency of FL training is heavily dependent on the quality of communication, given the continuous exchange of large volumes of model parameters. However, existing FL systems solely rely on terrestrial networks, which are increasingly strained by rising user numbers and service demands [8,9]. This often results in delays or congestion in parameter transmission [10], significantly affecting FL convergence, especially in remote or underdeveloped regions with poor signal coverage [11].

Fortunately, the proliferation of low-Earth-orbit (LEO) satellites is revolutionizing satellite broadband Internet [12,13], providing a crucial solution to the limitations of traditional terrestrial FL. In particular, SpaceX plans to deploy about 42 000 LEO satellites for its Starlink constellation [14]. LEO satellite networks offer broad coverage, facilitating peer-to-peer communication among numerous edge devices [15]. For instance, when devices involved

* Corresponding author.
E-mail address: gao.yue@fudan.edu.cn (Y. Gao).

https://doi.org/10.1016/j.eng.2025.07.020
2095-8099/© 2025 THE AUTHORS. Published by Elsevier LTD on behalf of Chinese Academy of Engineering and Higher Education Press Limited Company.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

in FL training (e.g., ocean or wilderness monitoring stations) encounter unstable terrestrial connections with central servers, satellite connectivity enables them to quickly acquire knowledge from multiple peers, improving their local training.

However, despite its promise, FL within satellite-assisted terrestrial networks remains largely unexplored. This is a complex endeavor that requires the meticulous design of transmission modes tailored to the intrinsic characteristics of satellite networks, as well as the effective utilization of satellite transmissions to address the various performance challenges posed by system heterogeneity. Firstly, the limited contact time and bandwidth of satellite networks—particularly in the uplink—significantly constrain the delivery of parameters. Secondly, ground devices typically operate in highly diverse environments, characterized by data heterogeneity, varying terrestrial network conditions, and differing local computing power. These system heterogeneities notably degrade model performance and present a primary challenge that satellite assistance must effectively address. Fig. 1 illustrates the satellite-assisted FL paradigm and its main challenges, which we will explore in greater detail in the next section.

To address these challenges, we propose a satellite-assisted FL framework for resource-constrained and heterogeneous environments, named SatFed. In SatFed, each edge device trains a personalized model for local inference while also contributing to global model training. Global model updates occur asynchronously between devices and the server via terrestrial networks, while devices exchange personalized models peer-to-peer through satellite networks for mutual guidance. SatFed utilizes freshness-priority queues for parameter transmission over bandwidth-constrained satellite networks, ensuring that devices receive the most up-to-date peer models. Additionally, a multigraph with three edge types is employed to dynamically capture diverse heterogeneities between devices, optimizing the local utilization of peer models received from satellite networks. The three edge types are ① similarity edges, which help devices learn from peers with similar data distributions; ② connection edges, which enable devices to benefit from peers with superior terrestrial bandwidth; and ③ computation edges, which monitor differences in device model update speeds and adjust the local learning rate accordingly. Our key contributions can be summarized as follows:

- To our knowledge, SatFed is a pioneering effort to harness LEO satellite networks for enhancing FL within traditional terrestrial networks. It tackles the distinctive transmission characteristics of satellite networks by implementing model freshness queues, effectively mitigating the challenges posed by constrained bandwidth on satellite-assisted performance.
- We investigate innovative solutions for addressing system heterogeneity in satellite-assisted terrestrial communication frameworks, employing a multigraph to monitor real-time model transmissions within satellite networks and capture intricate system heterogeneities. Through meticulous design of local updates, the multigraph-guided integration of satellite-transmitted models markedly improves training performance.
- We implement the SatFed prototype using actual LEO satellite networks and datasets. Comprehensive experiments demonstrate its capacity to achieve efficient and robust FL with satellite assistance across various heterogeneous conditions.

2. Motivation

In this section, we present empirical measurements that underscore the challenges inherent in satellite-assisted FL, thereby strengthening the motivation for the design of SatFed.

2.1. Constrained satellite network transmission

LEO satellites orbit periodically, engaging in bidirectional data transmission when within range of ground devices, and are primarily constrained by their brief contact times and limited uplink bandwidth. For instance, satellites orbiting at an altitude of 700 km complete one orbit approximately every 100 min, with contact windows to ground devices lasting only about 10 min [16]. As illustrated in Fig. 2(a), we assess parameter-transmission efficiency during these fleeting contact periods using the Starlink LEO satellite communication system. Fig. 2(b) presents the cumulative distribution function (CDF) derived from Iperf [17], a widely used

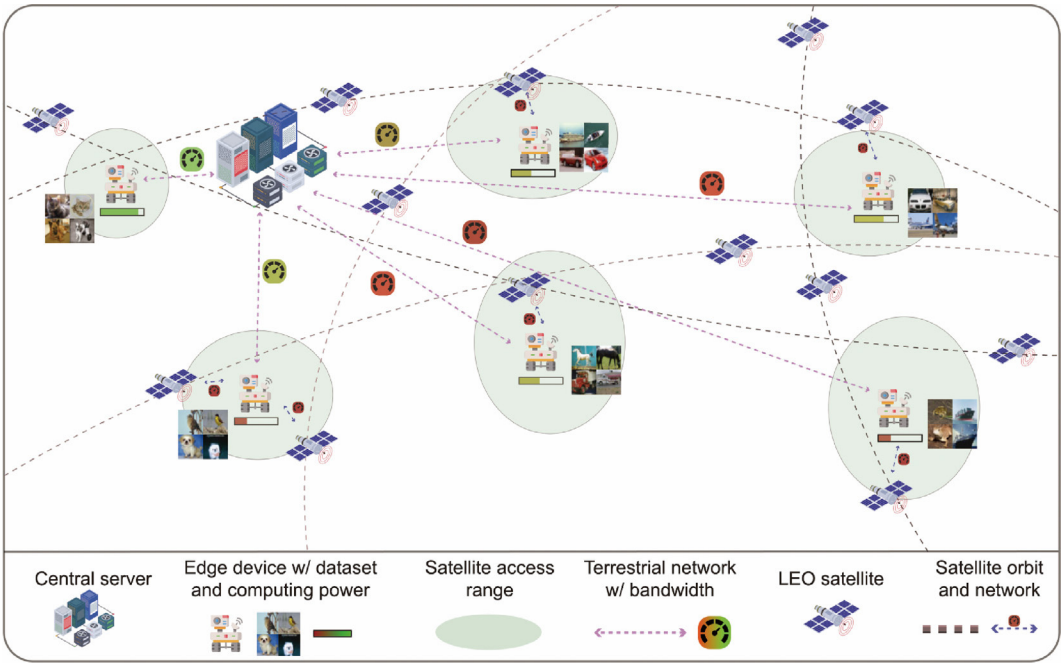


Fig. 1. While satellite networks offer a potential solution to the congestion and coverage limitations of traditional terrestrial FL, their efficiency presents major challenges from inherent satellite network constraints and FL system heterogeneity. w/: with.

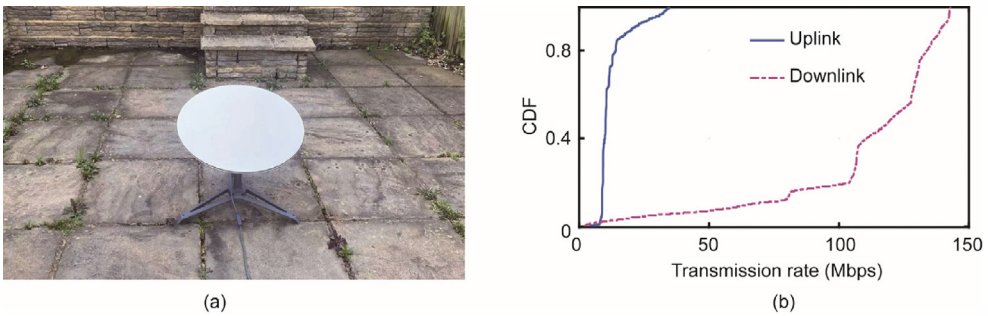


Fig. 2. The uplink in the LEO satellite network represents a significant transmission bottleneck. (a) Our experimental setup; (b) the resulting cumulative distribution function (CDF) showing the satellite network speed. Mbps: megabits per second.

network measurement tool, revealing an average downlink rate of approximately 100 megabits per second (Mbps) and an uplink rate of around 12 Mbps. For example, downloading a 98 Mb ResNet-50 model [18] from the satellite to each device takes about 8 s, whereas uploading the model back to the satellite requires more than a minute. Thus, the principal bottleneck in satellite network transmission lies in the uplink direction.

In an ideal scenario, each connection between a device and a satellite involves extensive model exchanges (as detailed further in Section 3.2). However, due to limitations in uplink transmission rates, the successful transmission of models (typically fewer than 10 in our ResNet-50 case) is significantly constrained. Thus, prioritizing critical models for satellite transmission is essential to optimize satellite-assisted performance.

2.2. Data heterogeneity

The performance of satellite-assisted FL is significantly impacted by data heterogeneity, as devices distributed globally encounter data from diverse contexts. These devices exhibit non-independent and non-identically distributed (non-IID) data, resulting in varied local optimization objectives [19]. Consequently, model updates from different devices are biased toward their local data distributions. Directly averaging these updates—as is done in the widely used FL method FedAvg [7]—often leads to suboptimal performance. To better understand the impact of data heterogeneity on training, we conducted experiments with the ResNet-50 model using FedAvg at varying levels of heterogeneity, as shown in Fig. 3(a) [20,21]. The results clearly indicate that performance degrades significantly as data heterogeneity increases.

Moreover, relying solely on a single global model can lead to fairness challenges across devices. To investigate this issue, we evaluated each device's local test accuracy under various heterogeneous conditions, as shown in Fig. 3(b) [20,21] (where the dashed line represents the average). It is evident that, as heterogeneity increases, the global model exhibits greater performance variance among devices, significantly reducing fairness. Although extensive research has been conducted on the impact of data heterogeneity on FL, most studies focus on PS communication frameworks [22] and do not adapt to emerging hybrid satellite–terrestrial network architectures.

2.3. Bandwidth heterogeneity

Terrestrial networks generally provide higher bandwidth than satellite networks, but they are subject to variability and heterogeneity due to factors such as fluctuating user volumes, routing conditions, and regional infrastructure disparities [23]. Devices operating in remote areas or congested environments (a.k.a., stragglers) experience longer delays in parameter transmission, which in turn slows down their local updates and ultimately hampers training performance. To better understand the impact of bandwidth heterogeneity on FL, we conducted experiments with varying proportions of stragglers. Fig. 4(a) shows that, as the proportion of stragglers increases, FL experiences significantly slower convergence and a marked deterioration in accuracy. As shown in Fig. 4(b), excluding outdated model updates from aggregation partially alleviates performance degradation but still falls short of the ideal. This comparison emphasizes that up-to-date models effectively extract local knowledge, while outdated models can negatively impact the system's performance.

Fortunately, the numerous satellites and extensive coverage in LEO constellations can enable edge devices to acquire timely knowledge from multiple peers during periods of ground network congestion or disconnection, thereby mitigating the issue of outdated local updates. However, effective design to address the straggler problem in such a specialized and hybrid network architecture remains largely understudied.

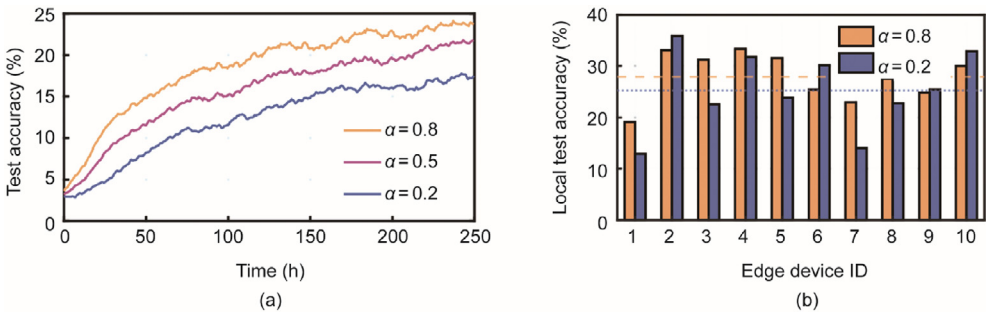


Fig. 3. The impact of data heterogeneity. (a) Heterogeneity levels; (b) fairness. The CIFAR-100 dataset [20] is distributed among ten edge devices, with local labels following a Dirichlet distribution; a smaller value of α indicates greater heterogeneity [21]. The experimental parameters are consistent with Section 4. ID: identification; α: a balancing factor.

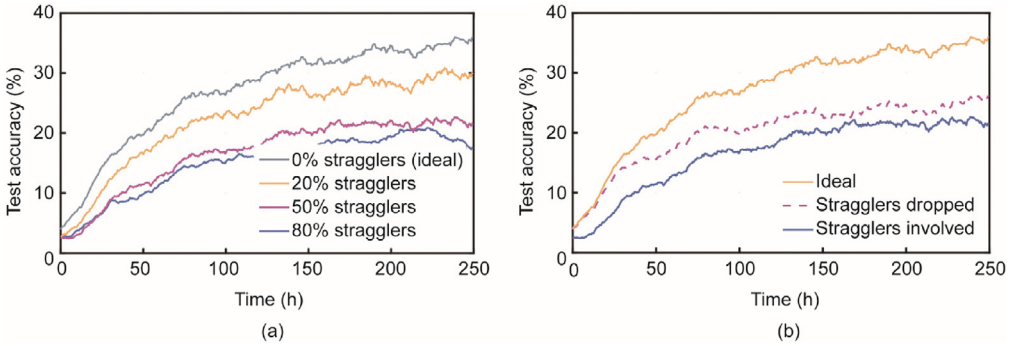


Fig. 4. The impact of bandwidth stragglers. (a) Straggler ratios; (b) damage from stragglers. Local updates were asynchronously aggregated to form and evaluate the global model. Stragglers have a 90% chance of communication blockage or disconnection with the server, retrying every 30 min.

2.4. Computation heterogeneity

Another challenge in satellite-assisted FL arises from the difference in computing capabilities among devices [19,24]. Certain devices are undertrained due to insufficient computational power within the same local update cycle. We conducted experiments to investigate how this uneven training in local updates across devices affects FL. As shown in Fig. 5(a) [25], the test accuracy of the global model significantly decreases as the number of undertrained devices increases. It should be noted that, as illustrated in Fig. 5(b) [25], the imbalance in the model updates resulting from computation heterogeneity can be mitigated by raising the learning rate for undertrained devices. Increasing the learning rates for normal devices does not improve performance, emphasizing that the improvement in performance comes from balancing update scales.

These findings show that the peer-to-peer communication mode of satellite networks is conducive to edge devices promptly perceiving the model update scales of their peers and adjusting local learning rates accordingly to ensure that all edge devices update with relatively consistent scales. However, in a hybrid satellite-assisted FL framework, strategies for the real-time and precise perception of peer update scales and the corresponding adjustment of local learning rates to achieve optimal performance require careful design.

3. Framework design

In this section, we elaborate on the SatFed framework and its well-designed components.

3.1. Preliminaries and overview

We consider a typical scenario of a LEO-satellite-assisted FL. Each edge device $i \in [m]$ possesses its own private dataset $\mathscr{D}_i$ from distribution $\mathbb{P}_i(x, y)$, where x and y denote the input features and corresponding labels, and m is the number of devices. Devices share a model ω^*. However, as outlined in Section 2.2, this model cannot effectively accommodate the diverse local data distributions of all devices. Therefore, we utilize personalized device models $\{v_i\}_{i\in[m]}$ for local inference on device. The optimization objective follows state-of-the-art FL methods using personalized models [26], where all devices are trained within the FedAvg framework to update the global model ω^* while using it to guide their own personalized model updates:

$$\min_{\{v_i\}} \frac{1}{m}\sum_{i=1}^{m}\left\{\mathscr{L}_i(v_i) + \frac{\mu}{2}\mathrm{Regu}(v_i, \omega^*)\right\} \tag{1}$$

$$\text{s.t.} \quad \omega^* \in \arg\min_{\omega} \sum_{i=1}^{m} \mathscr{L}_i(\omega) \tag{2}$$

where $\mathscr{L}_i(\omega) = \mathbb{E}_{(x,y)\sim\mathscr{D}_i}\mathscr{L}(f(\omega; x); y)$ is the empirical loss $\mathscr{L}$ of model ω (applicable to both models v_i and ω^*) on dataset $\mathscr{D}_i$, $f(\omega; x)$ denotes the output of model ω given input x, μ is a hyperparameter, and $\mathbb{E}$ denotes the expectation operator. $\mathrm{Regu}(\cdot,\cdot)$ regulates the parameter distance between the personalized model and the global model, achieving effective joint training. It is widely adopted in state-of-the-art personalized FL methods [26,27].

In SatFed, local updates of the global model are asynchronously transmitted to the server for aggregation via the terrestrial network. Simultaneously, personalized models are transferred peer to peer through the satellite network in a decentralized manner.

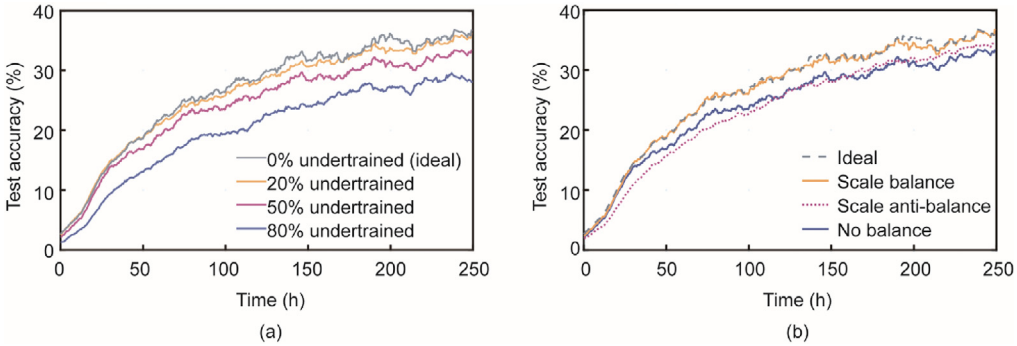


Fig. 5. The impact of uneven training. (a) Undertrained ratios; (b) balance effect. The local update period is set to 30 min, with normal devices having a compute power of 4 giga floating-point operations per second (GFlops; performance of a Raspberry Pi 4 Model B [25]), while undertrained devices have only 1/5 of that. The model is ResNet-50 with a batch size of 128. Scale balance (scale anti-balance) increases the local learning rates of undertrained (normal) devices by 1.5×.

The personalized model training is guided by both the global model and the peers' personalized models.

SatFed employs a model freshness-based priority-transmission mode, enabling edge devices to obtain more up-to-date models from peers in bandwidth-limited satellite networks. In addition, SatFed employs a multigraph that captures multiple heterogeneity relationships in real time using models transmitted in the satellite network. This in turn aids in distilling knowledge from cached peer models to ensure the performance and robustness of local updates. A system overview of the SatFed framework is illustrated in Fig. 6.

3.2. Satellite network transmission

In the transmission of a satellite network with K LEO satellites, satellite $k \in [K]$ (where K is the number of satellites), and device i maintain model caches $\mathscr{C}_{\text{sat}}^{k}$ and $\mathscr{C}^{i}$, respectively, for storing personalized models from various devices. As depicted in Fig. 6, when a device connects to a LEO satellite, it uploads models via the uplink. The satellite then transmits cached models to edge devices through the downlink as it passes overhead. However, the intricate topology and stringent bandwidth constraints of satellite networks, as discussed in Section 2.1, present significant challenges to naive satellite transmission—that is, to directly incorporating satellite communication into FL without any targeted design.

3.2.1. Drawbacks of naive transmission

Given L LEO satellite orbits with multiple satellites operating on each, for each orbit $l \in [L]$ (where L is the number of the satellite orbits), the set $S_l \subseteq [m]$ represents the edge devices it covers. When two edge devices i and j are simultaneously covered by any orbit—that is, $\exists l \in [L]$, $i \in S_l$, and $j \in S_l$—we term them "satellite-network direct," which is denoted as $\text{SatDir}_{i,j} = 1$; otherwise, the denotation is $\text{SatDir}_{i,j} = 0$. Notably, any device i may also be simultaneously covered by multiple orbits, which is denoted as the orbit set S_{orb}^{i}. It can be inferred that $S_i^{\text{SatDir}} = \{j | j \in m, \text{SatDir}_{i,j} = 1\}$ is the union $\bigcup \left\{ S_l | l \in S_{\text{orb}}^{i} \right\}$.

We consider two naive transmission modes: When device i connects to satellite k, it either ① uploads only its personalized device model v_i or ② additionally uploads all cached peer models $\{v_j | v_j \in \mathscr{C}^{i}\}$. The former method restricts satellite k in orbit l to caching only $\{v_i | i \in S_l\}$, which limits device j to receiving models solely from devices in S_j^{SatDir}, thereby causing transmission bias. As shown in Fig. 7(a), even with an average LEO satellite orbit covering up to ten devices and increasing the number of orbits to 50, devices may receive fewer than half of the peer models, resulting in notable bias in collaborative knowledge distribution. The second approach eliminates this bias by allowing satellites to receive models from other orbits. However, it significantly increases the transmission load on the uplink of the satellite network and introduces a substantial number of redundant transmissions. When the same model is transmitted to another device through multiple orbits, redundant transmission occurs, consuming the highly limited bandwidth resources of the satellite network. As shown in Fig. 7(b), as the number of orbits and coverage increases, this redundancy becomes increasingly severe, wasting tens of times the bandwidth resources.

3.2.2. Model freshness and priority queues

To tackle transmission bias and redundancy, SatFed implements model-priority queues to prioritize the transmission of critical models within the constrained bandwidth of satellite networks. Initially, we define the freshness of each model circulating in the satellite network to indicate its level of obsolescence. Specifically, each time a device i uploads v_i to the satellite, it timestamps the model as v_i^t, where t represents the upload initiation time. At any current time $\hat{t}$, the freshness of v_i^t is denoted as $\mathscr{R}\left(v_i^t, \hat{t}\right) = 1 - \mathrm{e}^{-\zeta\left(\hat{t}-t\right)}$, indicating that the closer the upload time of the model t is to the current time $\hat{t}$, the higher its freshness is, where ζ is a hyperparameter controlling the outdated speed.

Based on freshness, device i constructs upload and download model-priority queues, $Q_{i,k}^{\text{U}}$ and $Q_{i,k}^{\text{D}}$, respectively, when connected to satellite k. At the connection establishment time t^*, device i compares the freshness $\mathscr{R}\left(v_j^{t^1}, t^*\right)$ and $\mathscr{R}\left(v_j^{t^2}, t^*\right)$ of cached peer models $v_j^{t^1} \in \mathscr{C}^{i}$ and $v_j^{t^2} \in \mathscr{C}_{\text{sat}}^{k}$ for all $j \in [m]$ (where t^1 and t^2 are both different moments before t^*): If $\mathscr{R}\left(v_i^{t^1}, t^*\right) > \mathscr{R}\left(v_i^{t^2}, t^*\right)$, it adds $v_j^{t^1}$ to the upload queue $Q_{i,k}^{\text{U}}$; otherwise, it adds $v_j^{t^2}$ to the

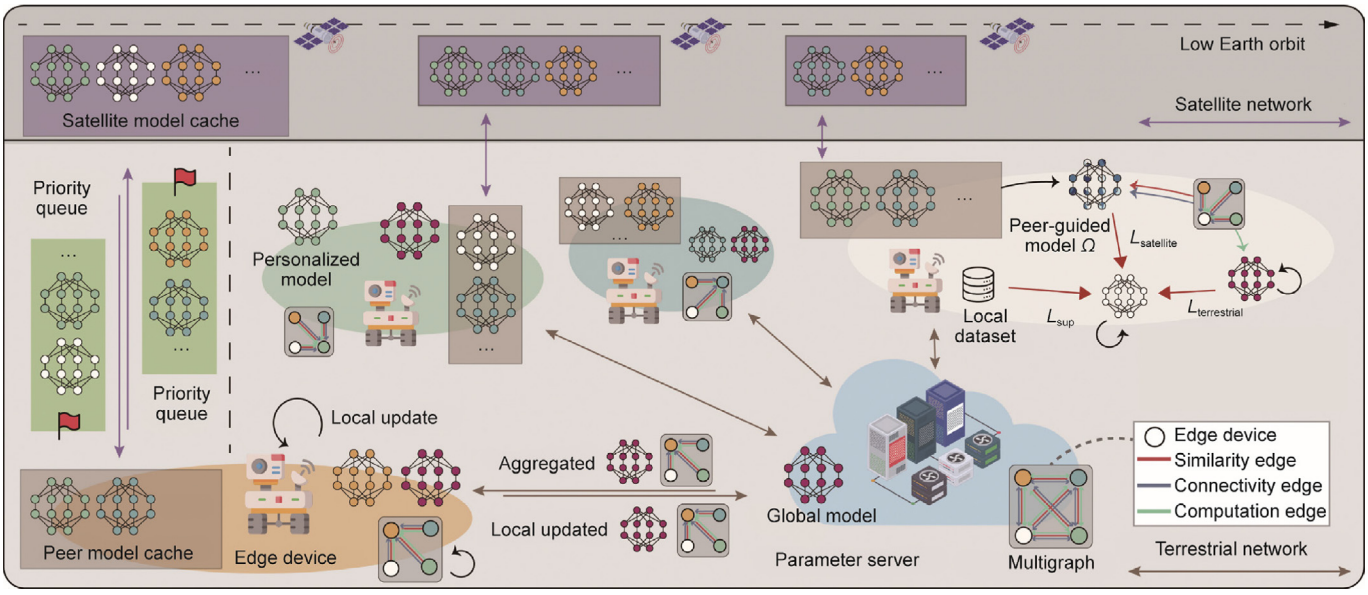


Fig. 6. System overview of the SatFed framework. L_{sup}: local supervised learning; $L_{\text{terrestrial}}$: global model guidance from the terrestrial network; $L_{\text{satellite}}$: peer model guidance from the satellite network.

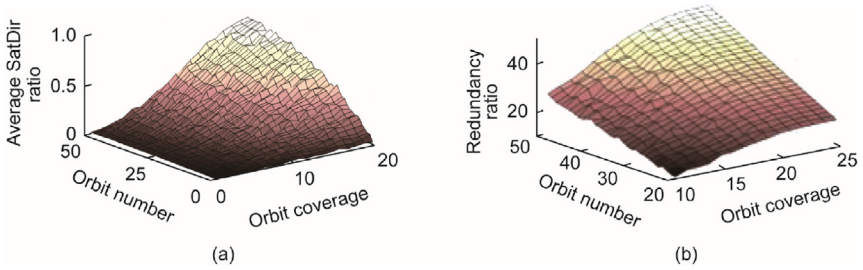


Fig. 7. Simulation of problems with naive satellite network transmission. (a) Transmission bias (average $|S_i^{\text{SatDir}}|/m$) across different topology configurations, where $m = 100$; (b) transmission redundancy when device i uploads the entire $\mathscr{C}^i$ upon connecting to the satellite. Redundancy indicates the ratio of redundant transmissions to the total required transmissions of transmitted models to necessary ones.

download queue $Q_{i,k}^{\text{D}}$. If a cache $\mathscr{C}$ lacks a model from device j, the cache is treated as $v_j^{-\infty} \in \mathscr{C}$. The models in the queues are arranged in descending order according to the freshness difference $|\mathscr{R}\left(v_i^{t^1}, t^*\right) - \mathscr{R}\left(v_i^{t^2}, t^*\right)|$. Following the priority sequence in the queues, device i subsequently uploads or downloads models, replacing the corresponding outdated models in the recipient cache.

Thus, SatFed tackles bias and redundancy in satellite network transmission via priority queues. ① If v_i does not exist in the recipient cache or is extremely outdated, the sender positions v_i at the front of the queue for priority transmission, thereby maximizing the alleviation of transmission bias. ② If two different satellites prepare to transmit personalized models from the same device with close upload times to the same recipient, the second model, having a minimal difference in freshness, is placed at the end of the transmission queue once the first model is received. The second model is transmitted only if the bandwidth is significantly idle, thus avoiding redundancy.

3.3. Heterogeneous multigraph

To address the multifaceted heterogeneity issues highlighted in Section 2, SatFed constructs a global multigraph, aiming to leverage model exchanges across satellite networks to capture the diverse relationships between devices and ultimately guide local updates. We document three types of heterogeneous relationships to construct the multigraph $\mathscr{G}_m = ([m], \mathscr{A}_{\text{sim}}, \mathscr{A}_{\text{con}}, \mathscr{A}_{\text{cmp}})$: ① similarity edges $\mathscr{A}_{\text{sim}}$; ② connection edges $\mathscr{A}_{\text{con}}$; and ③ computation edges $\mathscr{A}_{\text{cmp}}$. The details are as follows.

3.3.1. Similarity edges

As described in Section 2.2, diverse devices exhibit unique local empirical risks $\{\mathscr{L}_i\}_{i\in[m]}$ attributed to differences in training data distribution. Similarity edges $\mathscr{A}_{\text{sim}}$ aim to aid devices in discerning which peer models obtained from the satellite network better align with their optimization goals. Borrowing from state-of-the-art methods [28], we measure the similarity of local optimization objectives by calculating the cosine similarity of model parameters during training. However, in satellite networks where model freshness varies, simple cosine similarity may lead to inaccurate measurements, necessitating freshness calibration. Therefore, we compute the similarity edge $\mathscr{A}_{\text{sim}}$ between devices i and j with confidence $\kappa = \mathrm{e}^{-|t^1-t^2|}$ as follows:

$$\mathscr{A}_{\text{sim}}(i,j) = 1 - \frac{v_i^{t^1} \cdot v_j^{t^2}}{\left\|v_i^{t^1}\right\| \left\|v_j^{t^2}\right\|} \tag{3}$$

In short, when two models have similar freshness ($\kappa \to 1$), the cosine similarity accurately reflects the similarity of their local optimization objectives. Conversely, when their freshness differs significantly ($\kappa \to 0$), the cosine similarity fails to reflect their data distribution accurately. The confidence κ adjusts the similarity edge updates, as detailed in Section 3.3.4.

3.3.2. Connection edges

As discussed in Section 2.3, terrestrial bandwidth stragglers result in relatively outdated models. Connection edges help stragglers identify potential helpers. We monitor the communication frequency C_{S}^i between each edge device i and the PS over a specified time window W. Similarly, we record the frequency $C_{\text{D}}^{i,j}$ of successful transmissions of personalized models from devices j to i via the satellite networks. Subsequently, the connection edge is denoted as follows:

$$\mathscr{A}_{\text{con}}(i,j) = \frac{C_{\text{S}}^j + \delta \times C_{\text{D}}^{i,j}}{C_{\text{S}}^i} \tag{4}$$

where $\delta > 0$ is the weight balancing the two indicators. $\mathscr{A}_{\text{con}}(i,j)$ is large if device j has significantly more exchanges with the server than device i in a short period of time ($C_{\text{S}}^j \gg C_{\text{S}}^i$). This suggests that v_i can rely more on peer model v_j to keep up with the global training pace. Moreover, $C_{\text{D}}^{i,j}$ encourages v_i to learn from more familiar and easily accessible peer models, thereby reducing fluctuations in the convergence of the entire system.

3.3.3. Computation edges

To mitigate the global model degradation caused by the uneven training mentioned in Section 2.4, we devise the computation edges $\mathscr{A}_{\text{cmp}}$ to track the differences in update speeds across devices, thereby assisting in compensating for undertrained ones. We define $\mathscr{S}_i$ as the update speed of v_i as the Euclidean distance between parameters before and after the update, divided by the time interval (t' is the time when v_i was last updated):

$$\mathscr{S}_i = \frac{\left\|v_i^t - v_i^{t'}\right\|}{t - t'} \tag{5}$$

Subsequently, $\mathscr{A}_{\text{cmp}}$ is computed as the ratio of model update speeds between different devices:

$$\mathscr{A}_{\text{cmp}}(i,j) = \frac{\mathscr{S}_j}{\mathscr{S}_i} \tag{6}$$

where $\mathscr{S}_j$ is the update speed of v_j.

Based on $\mathscr{A}_{\text{cmp}}$, SatFed evaluates and compares the model update speeds of all edge devices and can adjust the local learning rate of undertrained models to prevent detrimental impacts on global model aggregation.

3.3.4. Multigraph update

The updates of the multigraph $\mathscr{G}_m$ are handled by both the server and the devices and are synchronized at the server side. At device i, updates occur at time t^* when downloading any peer model v_j^t via the satellite network, replacing the locally cached

old model $v_j^{t'}$. $\mathscr{A}_{\text{sim}}$ updates using weights based on the confidence mentioned earlier:

$$\mathscr{A}_{\text{sim}}(i,j) = \kappa \times \frac{v_i^{t^*} \cdot v_j^t}{\|v_i^{t^*}\| \|v_j^t\|} + (1-\kappa)\mathscr{A}_{\text{sim}}(i,j) \tag{7}$$

where confidence $\kappa = e^{-|t^*-t|}$. Additionally, device i recalculates the count of times it received v_j within the time window W to update $C_D^{i,j}$ and subsequently adjusts $\mathscr{A}_{\text{con}}(i,j)$ using Eq. (4). Finally, we compute $\mathscr{S}_j = \frac{\|v_j^t - v_j^{t'}\|}{t-t'}$ and similarly calculate $\mathscr{S}_i$, updating $\mathscr{A}_{\text{cmp}}(i,j)$ using Eq. (6).

3.4. Model training

This section elucidates how SatFed leverages the multigraph $\mathscr{G}_m$ to guide devices in extracting beneficial knowledge from transmitted models in satellite networks, thereby enhancing local updates and improving training performance.

3.4.1. Peer-guided model

In SatFed, devices cache many peer models from the satellite network, but these models differ greatly in their usefulness for local updates. Thus, using $\mathscr{G}_m$, we calculate the weight $\mathscr{T}(i,j)$, which represents device i's dependency on peer model v_j for local updates. This dependency conforms to two principles, in line with the design goals of $\mathscr{A}_{\text{sim}}$ and $\mathscr{A}_{\text{con}}$: ① The more similar their local optimization objectives, the stronger the mutual dependency; and ② the more frequent the recent communication between device j and the server, and the more often device j's model is sent to device i, the stronger i's dependency on j. Therefore, $\mathscr{T}(i,j)$ is defined as follows:

$$\mathscr{T}(i,j) = \mathscr{A}_{\text{sim}}(i,j) + \alpha \times \mathscr{A}_{\text{con}}(i,j) \tag{8}$$

where α is a balancing factor.

Using all peer model $v_j^{t'}$ in cache $\mathscr{C}^i$, combined with the respective dependency $\mathscr{T}(i,j)$, we generate a peer-guided model Ω_i for device i:

$$\Omega_i = \frac{1}{|\mathscr{C}^i|} \sum_{v_j^{t'} \in \mathscr{C}^i} e^{\hat{t}-t'} \times \mathscr{T}(i,j) \times v_j^{t'} \tag{9}$$

where $|\mathscr{C}^i|$ is the cache size, and the exponential term diminishes the impact of outdated models in the cache.

In short, Ω_i represents a synthesis of peer models that are ahead in global update progress and closely aligned with the optimization goals of v_i.

3.4.2. Global model update

In SatFed, the global model ω^* is locally trained by devices, with model parameters being transmitted via the terrestrial network to a PS for asynchronous updates. However, naive local updates result in the uneven training mentioned in Section 2.4, significantly affecting model performance. We address this issue by using $\mathscr{A}_{\text{cmp}}$ to dynamically adjust the local learning rate based on the real-time peer model-update speeds. More specifically, we use $U^i = \sum_{j\in[m]} \mathscr{A}_{\text{cmp}}(i,j)/m$ to denote the average update scale of peers for each device i. If $U^i \gg 1$, it indicates that device i is globally lagging behind, and its learning rate needs to be increased appropriately for better global model aggregation.

Thus, SatFed establishes a dynamic adaptive learning rate (η^i) for device i:

$$\eta^i = \eta \times \left(1 + \gamma \times \ln \max\left(1, U^i\right)\right) \tag{10}$$

where γ represents the gain coefficient. If the updating speed of v_i outpaces the global ($U_i \in (0,1)$), device i's learning rate η^i remains the preset hyperparameter η; otherwise, η^i is logarithmically increased with U_i. After each round of local updates begins, device i receives the latest global model ω^* from the server. It uses a learning rate η^i to train for R rounds of gradient descent on its dataset $\mathscr{D}_i$, obtaining the updated ω_i^*. The updated model is then sent back to the server for asynchronous aggregation with a fixed weight β.

3.4.3. Personalized model updates

The training of the personalized device model v_i is guided by both the global model ω^* and the peer-guided model Ω_i. It should be noted that, due to satellite assistance, local updates of v_i no longer need to wait for ω^* transmission and can proceed continuously. The local objective for v_i is as follows:

$$\min_{\{v_i\}} \frac{1}{m}\sum_{i=1}^{m} \left\{ \underbrace{\mathscr{L}_i(v_i)}_{L_{\text{sup}}} + \frac{\mu}{2} \underbrace{\|v_i - \omega^*\|_2^2}_{L_{\text{terrestrial}}} + \frac{\lambda}{2} \underbrace{\|v_i - \Omega_i\|_2^2}_{L_{\text{satellite}}} \right\} \tag{11}$$

where the optimization objective consists of three components—local supervised learning (L_{sup}), global model guidance from the terrestrial network ($L_{\text{terrestrial}}$), and peer model guidance from the satellite network ($L_{\text{satellite}}$). λ is a hyperparameter.

The pseudocode details of SatFed are provided in Algorithm 1. Obviously, satellite transmission allows devices to promptly receive and evaluate model updates from peers, facilitating local updates and effectively mitigating the negative impact of system heterogeneity on FL training. Without satellite communication, devices with limited terrestrial bandwidth or computational power severely degrade FL performance, as discussed in detail in Section 4.3. Notably, while SatFed is tailored for terrestrial and satellite networks, it offers insights into hybrid centralized/decentralized FL frameworks in general.

Algorithm 1. SatFed.

Input: η, R, β, μ, and λ
Output: $\{v_i\}_{i\in[m]}$
Initialization: $\mathscr{G}_m$, $\{v_i\}_{i\in[m]}$, ω^*
Server executes:
for each device $i \in [m]$ **asynchronously do**
  Server sends ω^*, $\mathscr{G}_m$ to device i {terrestrial}
  Device i updates $\mathscr{C}^i$, $\mathscr{G}_m$ {satellite}
  $\omega_i^* \leftarrow$ LocalUpdate (v_i, ω^*, $\mathscr{C}^i$, $\mathscr{G}_m$, μ, λ, η)
  Device i sends ω_i^*, $\mathscr{G}_m$ back {terrestrial}
  $\omega^* = \beta \times \omega_i^* + (1-\beta) \times \omega^*$
  Server updates $\mathscr{G}_m$ based on Section 3.3.4
end for
LocalUpdate (v_i, ω^*, $\mathscr{C}^i$, $\mathscr{G}_m$, μ, λ, η)
Build Ω_i based on Section 3.4.1
Compute η^i based on Section 3.4.1
$\omega_i^* = \omega^*$
for each local epoch r = 1, 2, ...,R **do**
  Randomly selects a batch $\mathscr{B}_i \in \mathscr{D}_i$
  $\omega_i^* = \omega_i^* - \eta^i \nabla \mathscr{L}_i(\omega_i^*; \mathscr{B}_i)$
  $v_i = v_i - \eta \nabla \mathscr{L}_i(v_i; \mathscr{B}_i) - \eta\mu(v_i - \omega_i^*) - \eta\lambda(v_i - \Omega_i)$
end for
return ω_i^*

4. Performance evaluation

This section presents numerical results to evaluate the training performance of the proposed SatFed framework and the effectiveness of each individual component.

4.1. Experimental setup

We implemented SatFed using Python 3.7 and PyTorch 1.12.1 and trained it with NVIDIA GeForce RTX 3090 graphics processing units (GPUs). In the experiments, $m = 20$ edge devices were deployed, along with $K = 20$ satellites orbiting the Earth, distributed across $L = 10$ LEO orbits. We used two widely acclaimed image datasets, Fashion-MNIST [29] and CIFAR-100 [21], utilizing a Dirichlet distribution to generate non-IID label distributions for the devices [22], and set the α parameter to 0.2 for strongly non-IID settings and to 0.5 for relatively independent identically distribution (IID) settings. The data quantities per device were randomly assigned, ranging from 1500 to 2000. We employed a ResNet-50 model with a size of 98 Mb for CIFAR-100 (ResNet-18 for Fashion-MNIST). An Adam optimizer [30] was used, with the learning rate η set to 0.001 and the batch size set to 128. The loss function $\mathscr{L}_i$ represented the cross-entropy loss.

According to real LEO satellite networks [16], each satellite orbit had an operational cycle of 100 min and covered approximately three edge devices, with a contact time of about 10 min per device. The bandwidth of the satellite network was set to 100 Mbps for the downlink and 10 Mbps for the uplink as measured. The local update period was set to 30 min. Half the devices had a compute power of 4 GFlops (equivalent to a Raspberry Pi 4 Model B), while the others had only 1/5 of that. Half the devices typically operated under an ideal 200 Mbps terrestrial network bandwidth, while the other half had a 90% chance of communication blockage or disconnecting from the server, retrying every 30 min.

4.2. Overall performance

In this section, we evaluate the overall performance of the SatFed framework in terms of test accuracy and convergence speed. In addition, we assess the impact of satellite network transmission on SatFed's training. To investigate the advantages of SatFed, we compare it with five other benchmarks:

- **FedAvg:** The server waits for all devices to return their model updates, aggregates them into a new global model, and sends it to devices to start the next round [7].
- **FedAsync:** The updated models from all devices are aggregated into the latest global model in an asynchronous manner [31].
- **Ditto:** Each device maintains a personalized model, trains a global model using FedAvg, and incorporates it to aid personalized model updates with L2 regularization [26].
- **Ditto-Async:** This asynchronous variant of Ditto updates the global model following FedAsync.
- **SatFed(−):** SatFed's continuous local update method is employed without incorporating a peer guide model during training.

4.2.1. The overall performance of SatFed

Fig. 8 presents the test accuracy of SatFed compared with other benchmarks on the Fashion-MNIST and CIFAR-100 datasets. FedAvg and Ditto exhibit significantly slower convergence due to waiting for bandwidth-straggling devices during each global model aggregation. In contrast, FedAsync and Ditto-Async accelerate convergence at the cost of training instability, which can be attributed to the degradation caused by outdated models from asynchronous updates. It is evident that SatFed achieves the fastest convergence among all benchmarks, facilitated by its connection edges, which enable stragglers to learn from high-bandwidth peers via satellite networks and thereby alleviate local model drag on the system convergence progress. Moreover, SatFed achieves significantly higher test accuracy than the other benchmarks, thanks to its similarity edges, which enable devices to selectively collaborate with peers that have similar data distributions. Furthermore, the computation edges adapt to differences in model update speeds, guiding the adjustment of local learning rates to mitigate the negative impact of uneven training on the global model. Notably, SatFed(−) does not achieve the same performance as SatFed, suggesting that peer guidance is the key to SatFed's superior performance.

Table 1 presents the converged accuracies and speeds for the Fashion-MNIST and CIFAR-100 datasets. It is evident that SatFed achieves higher test accuracies of 2.35% and 2.26% (1.07% and 1.01%) in the IID and non-IID settings on CIFAR-100 (Fashion-MNIST) compared with the other methods, with convergence speeds up to 100× faster than those of FedAvg. SatFed's fast convergence is mainly due to its satellite-assisted guidance from high-bandwidth devices, which prevents personalized models from becoming outdated. Compared with FedAvg (FedAsync), Ditto (Ditto-Async) achieves better test accuracy in non-IID environments, with an average improvement of 8.12% due to the use of personalized models. Compared with Ditto-Async, SatFed effectively enhances collaboration among similar peers, while also sensing and compensating for uneven training, further improving performance.

4.2.2. The impact of satellite network transmission

Fig. 9 shows the impact of different satellite network transmission modes on SatFed's training performance. As described in Section 3.2.1, if devices upload only their own models, they can only receive models from their satellite-network direct peers, leading to biases that undermine the effectiveness of peer-guided models. In contrast, the redundant transmission mode involves devices uploading all cached models to satellites, thereby mitigating bias in peer-guided model knowledge sources. However, this results in a surge in transmission loads, overwhelming the highly constrained uplink links and causing massive model-upload failures. It is evident that SatFed's use of a freshness-priority model queue leads to the best performance. This is because the queue prioritizes the transmission of more critical models based on freshness in constrained satellite networks, ensuring overall model currency across the satellite networks. Consequently, peer-guidance knowledge remains up to date, resulting in training-performance improvements.

4.3. Micro-benchmark

4.3.1. Similarity edges

Fig. 10 illustrates the effect of similarity edges on the training of the CIFAR-100 dataset. Clearly, SatFed with $\mathscr{A}_{\text{sim}}$ achieves optimal performance in both IID and non-IID settings. Remarkably, SatFed demonstrates performance improvements over Ditto even without similarity edges. This result suggests that using the simply averaged peer models from satellite networks as the peer-guided model Ω is still beneficial, albeit to a modest extent. This improvement is more significant in the non-IID scenario, reaching 1.17%, whereas it is negligible in the IID scenario. This result is likely because, in the IID setting, the global model excels in knowledge extraction, whereas its efficacy diminishes in the non-IID scenario, necessitating supplementary knowledge from peers. A comparison of SatFed's accuracy with and without $\mathscr{A}_{\text{sim}}$ shows that the similarity capture in model transmissions within satellite networks boosts performance by 2.34% in IID and 1.92% in non-IID scenarios. By mitigating the impact of outdated models on similarity updates and effectively guiding Ω, $\mathscr{A}_{\text{sim}}$ significantly improves the performance of local updates.

4.3.2. Connection edges

Fig. 11 presents the impact of connection edges on CIFAR-100 dataset training performance. A comparison of the training

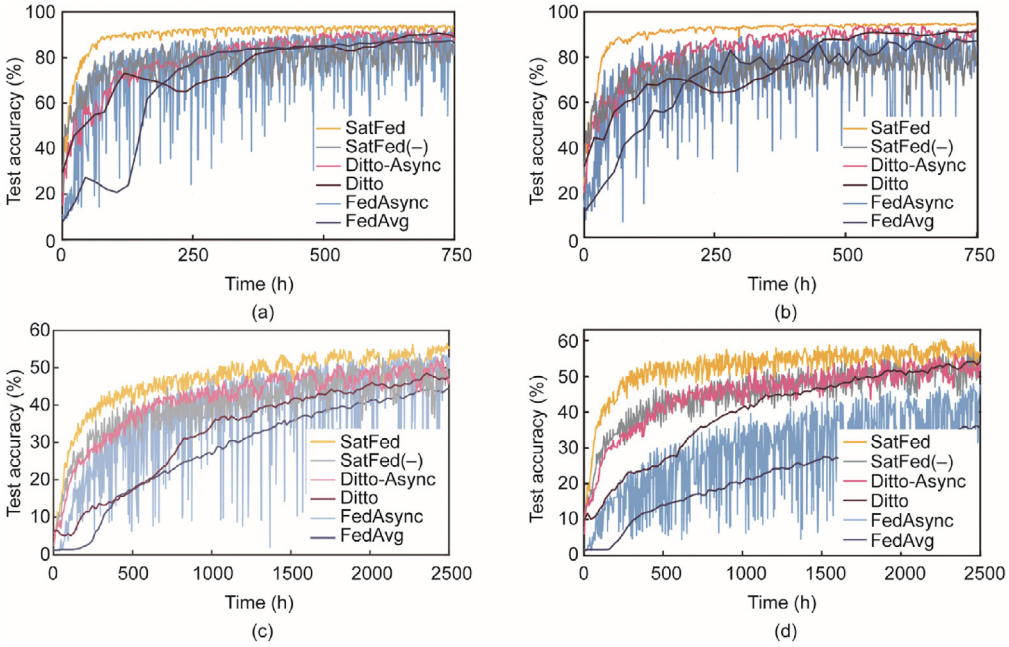


Fig. 8. Test accuracy for the Fashion-MNIST and CIFAR-100 datasets under IID and non-IID settings. (a) Fashion-MNIST (IID); (b) Fashion-MNIST (non-IID); (c) CIFAR-100 (IID); (d) CIFAR-100 (non-IID).

Table 1
Converged test accuracy (with device standard deviation) and speed of SatFed and other benchmarks.

Method	Converged accuracy (%) and speed on the Fashion-MNIST dataset		Converged accuracy (%) and speed on the CIFAR-100 dataset	
	$\alpha = 0.2$ (non-IID)	$\alpha = 0.5$ (IID)	$\alpha = 0.2$ (non-IID)	$\alpha = 0.5$ (IID)
FedAvg	91.75 ± 5.07	91.26 ± 4.65	50.30 ± 4.39	54.57 ± 2.91
FedAsync	92.48 ± 4.67 (5.66×)	92.07 ± 3.40 (3.39×)	48.34 ± 5.91 (12.89×)	54.20 ± 3.27 (12.18×)
Ditto	94.07 ± 4.50 (1.08×)	93.41 ± 3.87 (0.89×)	58.56 ± 5.16 (3.39×)	52.82 ± 3.17 (1.31×)
Ditto-Async	93.88 ± 4.94 (3.72×)	92.85 ± 3.47 (2.36×)	56.31 ± 7.09 (22.22×)	53.06 ± 4.32 (10.91×)
SatFed(–)	91.89 ± 7.08 (3.34×)	92.05 ± 3.77 (3.70×)	58.41 ± 8.14 (25.72×)	53.75 ± 3.92 (11.84×)
SatFed	**95.08** ± 3.02 (**13.82**×)	**94.48** ± 2.23 (**12.16**×)	**60.82** ± 5.23 (**100.60**×)	**56.92** ± 2.97 (**19.19**×)

The optimal values are in bold. Speed is the reciprocal of the ratio between the epochs needed to reach FedAvg's converged accuracy and the epochs required by FedAvg itself.

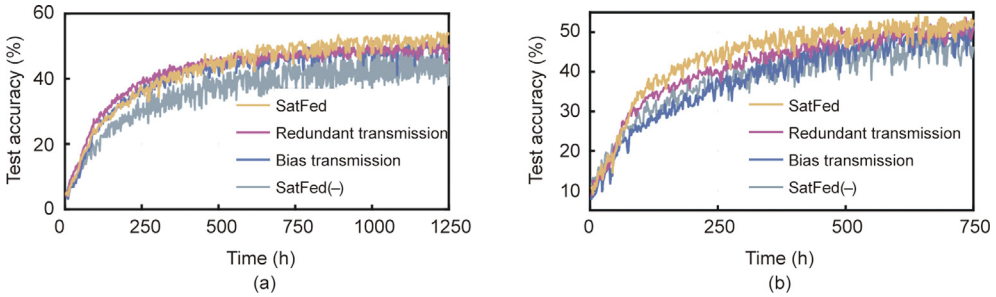


Fig. 9. The impact of satellite network transmission methods on training performance. (a) CIFAR-100 (IID); (b) CIFAR-100 (non-IID). Bias and redundant transmissions are detailed in Section 3.2.1; SatFed employs the freshness-priority queue scheme.

performance with and without $\mathscr{A}_{\text{con}}$ reveals that the edge connection's impact is most significant in the early stages of training. At this point, all devices' model updates are relatively similar. Models with poor terrestrial bandwidth can learn directly from the faster-progressing models through $\mathscr{A}_{\text{con}}$, significantly accelerating overall convergence. However, in the later stages, due to the impact of data heterogeneity, the update directions of different devices diverge significantly, reducing the effectiveness of $\mathscr{A}_{\text{con}}$.

4.3.3. Connection edges

Fig. 12 illustrates the effect of computation edges on SatFed training. Clearly, the assistance of $\mathscr{A}_{\text{cmp}}$ in increasing the learning rate of undertrained models significantly improves the performance of the aggregated global models in IID scenarios. Although the improvement in non-IID scenarios is less pronounced, it still effectively increases the injection of global knowledge into personalized models. In summary, computation edges effectively improve

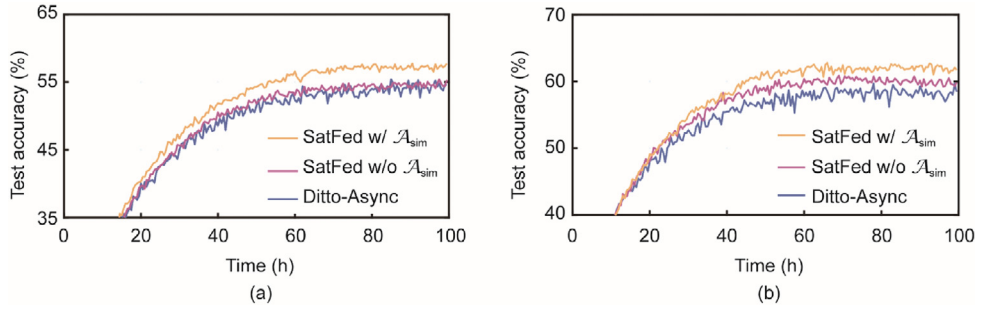


Fig. 10. Ablation experiments for similarity edges. (a) CIFAR-100 (IID); (b) CIFAR-100 (non-IID). To emphasize the impact of data heterogeneity, all edge devices are under ideal bandwidth and computational conditions. w/o: without.

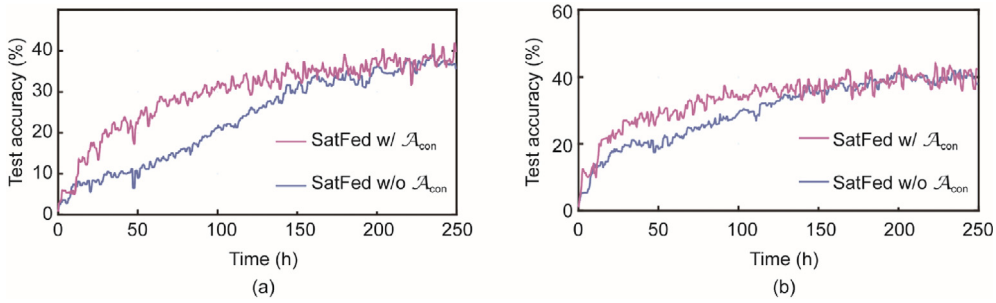


Fig. 11. Ablation experiments for connection edges. (a) CIFAR-100 (IID); (b) CIFAR-100 (non-IID). To highlight the terrestrial bandwidth heterogeneity, all devices train with ideal computing power.

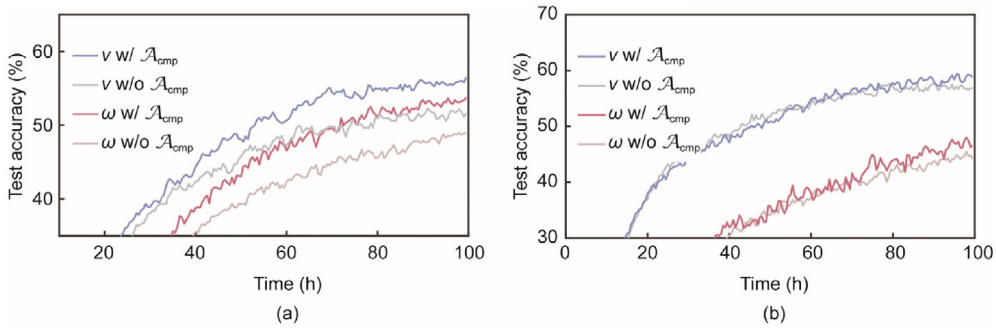


Fig. 12. Ablation experiments for computation edges. (a) CIFAR-100 (IID); (b) CIFAR-100 (non-IID). To emphasize the impact of computation heterogeneity, all devices operate within ideal terrestrial bandwidth.

SatFed's performance across both IID and non-IID settings. This result underscores the ability of computation edges to capture differences in local training progress and effectively address these differences through learning rate adjustments.

5. Related work

The system heterogeneity in FL, which encompasses data, bandwidth, and computation heterogeneity, is a well-established field with various approaches. To address data heterogeneity challenges, Li et al. [26] introduced personalized device models, increasing the fairness and robustness of the system. Smith et al. [32] adapted multitask learning for personalization by treating different devices as distinct learning tasks. Ye et al. [28] customized the aggregation weights for each device by measuring the similarity between personalized models. To tackle the issue of bandwidth heterogeneity, Nguyen et al. [33] maintained a buffer for the aggregation of device updates, reducing the performance degradation caused by outdated models. Wang et al. [34] designed a cluster-based FL mechanism with hierarchical aggregation, balancing communication resources by combining synchronous and asynchronous aggregations. As for the uneven training issues caused by computational heterogeneity, Diao et al. [24] deployed device-specific local models adapted to match local computational resources and designed their aggregation method accordingly.

However, existing methods that address FL system heterogeneities mainly focus on terrestrial PS communication architectures, overlooking the potential offered by emerging hybrid communication pathways such as satellite-assisted terrestrial networks. This paper proposes novel solutions utilizing LEO-satellite-assisted communication to tackle heterogeneity challenges, presenting an orthogonal approach compared with the aforementioned methods.

Satellite–terrestrial networks are an emerging and promising hybrid architecture with significant potential to alleviate terrestrial network transmission bottlenecks and improve overall transmission service quality [35]. Cui et al. [36] utilized satellite networks to assist terrestrial IoT networks in reducing

transmission latency. Ding et al. [37] designed dynamic transmission and computational resource optimization for dense LEO-satellite-assisted mobile-edge computing. Although LEO satellites show potential to assist in terrestrial communications, the exploration of FL in satellite–terrestrial hybrid network architectures remains largely uncharted. This problem is non-trivial, requiring the systematic design of model training, aggregation, and transmission for hybrid network architectures, while addressing satellite network topology, bandwidth issues, and various heterogeneity challenges in FL training.

6. Conclusions

In this paper, we proposed SatFed, a resource-efficient satellite-assisted FL framework designed to enhance training performance and convergence speed in complex heterogeneous environments. First, SatFed designs a priority transmission queue based on model freshness differences, enabling devices in satellite networks with highly constrained transmission rates and complex topologies to receive near-real-time peer models. Furthermore, SatFed utilizes a multigraph to perceive the heterogeneous relationships among devices in real time, as indicated by the models transmitted in satellite networks. Finally, SatFed utilizes perceived heterogeneity relationships by constructing peer guide models from satellite transmission, thus directing local updates. Extensive experiments showed that the SatFed framework outperforms state-of-the-art benchmarks. This work underscores the potential of deploying SatFed across real LEO-satellite-assisted terrestrial networks.

CRediT authorship contribution statement

Yuxin Zhang: Writing – original draft, Software, Methodology, Conceptualization. **Zheng Lin:** Writing – review & editing, Methodology. **Zhe Chen:** Writing – review & editing, Supervision, Conceptualization. **Zihan Fang:** Investigation, Data curation. **Xianhao Chen:** Supervision. **Wenjun Zhu:** Resources, Data curation. **Jin Zhao:** Project administration, Investigation. **Yue Gao:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (62341105).

References

[1] Liu J, Ren J, Zhang Y, Yue S, Zhang Y. SESAME: a resource expansion and sharing scheme for multiple edge services providers. IEEE/ACM Trans Netw 2024;32(4):3189–204.
[2] Zhang X, Xu H, Zou L, Duan J, Wu C, Xue Y, et al. Rosevin: employing resource- and rate-adaptive edge super-resolution for video streaming. In: Proceedings of the IEEE Annual Joint Conference: INFOCOM, IEEE Computer and Communications Societies; 2024 May 20–23; New York City, NY, USA. New York City: IEEE; 2024. p. 1–10.
[3] Ren J, Liu J, Zhang Y, Li Z, Lyu F, Wang Z, et al. An efficient two-layer task offloading scheme for MEC system with multiple services providers. In: Proceedings of IEEE Conference on Computer Communications; 2022 May 2–5; online. New York City: IEEE; 2022. p. 1519–28.
[4] Zheng T, Li A, Chen Z, Wang H, Luo J. AutoFed: heterogeneity-aware federated multimodal learning for robust autonomous driving. In: Proceedings of the 29th Annual International Conference on Mobile Computing and Networking; 2023 Oct 2–6; Madrid, Spain. New York City: Association for Computing Machinery (ACM); 2023. p. 1–15.
[5] Marelli L, Testa G. Scrutinizing the EU general data protection regulation. Science 2018;360(6388):496–8.
[6] Rodio A, Faticanti F, Marfoq O, Neglia G, Leonardi E. Federated learning under heterogeneous and correlated client availability. IEEE/ACM Trans Netw 2023;32(2):1451–60.
[7] McMahan B, Moore E, Ramage D, Hampson S, Arcas BA. Communication-efficient learning of deep networks from decentralized data. In: Proceedings of the 20th International Conference on Artificial Intelligence and Statistics; 2017 Apr 20–22; Ft. Lauderdale, FL, USA. New York City: MLResearchPress; 2017. p.1273–82.
[8] Sacco A, Flocco M, Esposito F, Marchetto G. Owl: congestion control with partially invisible networks via reinforcement learning. In: Proceedings of IEEE Conference on Computer Communications; 2021 May 10–13; online. New York City: IEEE; 2021. p. 1–10.
[9] Öhlén P, Skubic B, Rostami A, Fiorani M, Monti P, Ghebreten-saé Z, et al. Data plane and control architectures for 5G transport networks. J Lit Technol 2016;34(6):1501–8.
[10] Yang D, Zhang W, Ye Q, Zhang C, Zhang N, Huang C, et al. DetFed: dynamic resource scheduling for deterministic federated learning over time-sensitive networks. IEEE Trans Mobile Comput 2024;23(5):5162–78.
[11] Xu J, Kishk MA, Alouini MS. Space–air–ground–sea integrated networks: modeling and coverage analysis. IEEE Trans Wirel Commun 2023;22(9):6298–313.
[12] Zhai Z, Wu Q, Yu S, Li R, Zhang F, Chen X. FedLEO: an offloading-assisted decentralized federated learning framework for low Earth orbit satellite networks. IEEE Trans Mobile Comput 2024;23(5):5260–79.
[13] Zhang Y, Wu Q, Lai Z, Li H. Enabling low-latency-capable satellite–ground topology for emerging LEO satellite networks. In: Proceedings of IEEE Conference on Computer Communications; 2022 May 2–5; New York City, NY, USA. New York City: IEEE; 2022. p. 1329–38.
[14] Ahmmed T, Alidadi A, Zhang Z, Chaudhry AU, Yanikomeroglu H. The digital divide in Canada and the role of LEO satellites in bridging the gap. IEEE Commun Mag 2022;60(6):24–30.
[15] Lai Z, Li H, Wang Y, Wu Q, Deng Y, Liu J, et al. Achieving resilient and performance-guaranteed routing in space–terrestrial integrated networks. In: Proceedings of IEEE Conference on Computer Communications; 2023 May 17–20; New York City, NY, USA. New York City: IEEE; 2023. p. 1–10.
[16] Apollonio P, Caini C, Lülf M. DTN LEO satellite communications through ground stations and GEO relays. In: Dhaou R, Beylot AL, Montpetit MJ, Lucani D, Mucchi L, editors. Personal satellite services. Cham: Springer; 2013. p. 1–12.
[17] iPerf—the ultimate speed test tool for TCP, UDP and SCTP [Internet]. Reims: iKoula; [cited 2025 Jun 18]. Available from: https://iperf.fr/.
[18] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition; 2016 Jun 27–30; Las Vegas, NV, USA. New York City: IEEE; 2016. p. 70–8.
[19] Li T, Sahu AK, Zaheer M, Sanjabi M, Talwalkar A, Smith V. Federated optimization in heterogeneous networks. In: Proceedings of Machine Learning and Systems 2; 2020 Mar 2–4; Austin, TX, USA. Indio: MLSys Proceedings; 2020. p. 429–50.
[20] Chen S, Li B. Towards optimal multi-modal federated learning on non-IID data with hierarchical gradient blending. In: Proceedings of IEEE Conference on Computer Communications; 2022 May 2–5; London, UK. New York City: IEEE; 2022. p. 1469–78.
[21] Tan AZ, Yu H, Cui L, Yang Q. Towards personalized federated learning. IEEE Trans Neural Netw Learn Syst 2022;34(12):9587–603.
[22] Krizhevsky A, Hinton G. Learning multiple layers of features from tiny images. Toronto: University of Toronto; 2009.
[23] Liao Y, Xu Y, Xu H, Wang L, Qian C. Adaptive configuration for heterogeneous participants in decentralized federated learning. In: Proceedings of IEEE Conference on Computer Communications; 2023 May 17–20; Hoboken, NJ, USA. New York City: IEEE; 2023. p. 1–10.
[24] Diao E, Ding J, Tarokh V. HeteroFL: computation and communication efficient federated learning for heterogeneous clients. 2020. arXiv:2010.01264.
[25] Basford PJ, Johnston SJ, Perkins CS, Garnock-Jones T, Tso FP, Pezaros D, et al. Performance analysis of single board computer clusters. Future Gener Comput Syst 2020;102:278–91.
[26] Li T, Hu S, Beirami A, Smith V. Ditto: fair and robust federated learning through personalization. In: Proceedings of the 38th International Conference on Machine Learning; 2021 Jul 18–24; online. New York City: MLResearchPress; 2017. p.6357–68.
[27] Dinh TC, Tran NH, Nguyen TD. Personalized federated learning with moreau envelopes. In: Proceedings of the 34th International Conference on Neural Information Processing Systems; 2020 Dec 6–12; Vancouver, BC, Canada. Red Hook: Curran Associates Inc.; 2020. p. 21394–405.
[28] Ye R, Ni Z, Wu F, Chen S, Wang Y. Personalized federated learning with inferred collaboration graphs. In: Proceedings of 40th International Conference on Machine Learning; 2023 Jul 23–29; Honolulu, HI, USA. New York City: MLResearchPress; 2023. p. 39801–17.
[29] Xiao H, Rasul K, Vollgraf R. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms. 2017. arXiv:1708.07747.
[30] Kingma DP, Ba J. Adam: a method for stochastic optimization. 2014. arXiv:1412.6980.
[31] Xie C, Koyejo S, Gupta I. Asynchronous federated optimization. 2019. arXiv:1903.03934.

[32] Smith V, Chiang CK, Sanjabi M, Talwalkar AS. Federated multi-task learning. In: Proceedings of the 31st International Conference on Neural Information Processing Systems; 2014 Dec 4–9; Long Beach, CA, USA. Red Hook: Curran Associates Inc.; 2014. p. 4427–37.
[33] Nguyen J, Malik K, Zhan H, Yousefpour A, Rabbat M, Malek M, et al. Federated learning with buffered asynchronous aggregation. In: Proceedings of The 25th International Conference on Artificial Intelligence and Statistics; 2022 Mar 28–30; online. New York City: MLResearchPress; 2022. p.3581–607.
[34] Wang Z, Xu H, Liu J, Huang H, Qiao C, Zhao Y. Resource-efficient federated learning with hierarchical aggregation in edge computing. In: Proceedings of the IEEE Annual Joint Conference: INFOCOM, IEEE Computer and Communications Societies; 2021 May 10–13; online. New York City: IEEE; 2021. p. 1–10.
[35] Sun Y, Peng M, Zhang S, Lin G, Zhang P. Integrated satellite–terrestrial networks: architectures, key techniques, and experimental progress. IEEE Netw 2022;36(6):191–8.
[36] Cui G, Duan P, Xu L, Wang W. Latency optimization for hybrid GEO–LEO satellite-assisted IoT networks. IEEE Internet Things J 2023;10(7):6286–97.
[37] Ding C, Wang JB, Cheng M, Lin M, Cheng J. Dynamic transmission and computation resource optimization for dense LEO satellite assisted mobile-edge computing. IEEE Trans Commun 2023;71(5):3087–102.